Problem Set 1 Solutions

Course 18.06 Linear Algebra (MIT OCW, Spring 2020) · **Student** Songpengfei · **Date** 2026-10-09 · **Solved** 5/5 (100.0%)

Problem P1

Warmup questions (20 points, -4 for wrong answers until 0 is reached).
The following are supposed to be familiar from 18.02, but may not be. We recommend reading all of Strang chapter 1, but especially 1.2 for dot products. Also there is 18.02 week 1 course notes on OCW.

(a) $1)$ **unit vector**

(b) $2)$ **zero vector**

(c) a, b, c 都正确（**perpendicular, orthogonal, at right angles**）

(d) (a) $1)$ **unit vector;** (b) $2)$ **zero vector;** (c) $a), b), c)$; $(d2)$ **True;** (e) $2)$ $45°$; (f) $\det A = \pm 2$

(d2) **True**

(e) $45°$（选项 2）

(f) $\det A = \pm 2$

> **Answer**
> (a) $1)$ **unit vector** (b) $2)$ **zero vector** (c) a, b, c 都正确（**perpendicular, orthogonal, at right angles**） (d) (a) $1)$ **uni**

`mixed` unknown

Problem P2

Warmup questions (20 points, -4 for wrong answers until 0 is reached).
Another 18.02 concept is gradients as in the week 5 18.02 course notes on OCW.

(a) $3)$ **plane not through 0**

(b) $(2, 3, 4)$

(c) $d)$**sphere**

(d) (x, y, z)

(e) $\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

(f) $\nabla f = (2x, 2y, 2z)$

(g) 半空间 $2x + 3y + 4z < 2020$

> **Answer**
> (a) $3)$ **plane not through 0** (b) $(2, 3, 4)$ (c) $d)$**sphere** (d) (x, y, z) (e) $\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ (f) $\nabla f = (2x, 2y, 2z)$ (g) 半空间 $2x$

`mixed` calculus

Problem P3

Am I a (real)vector space? (20 pts) (A few criteria to be a vector space is that real linear combinations make sense, and remain inside the space. In particular there should be a zero in the space. Please answer yes or no and explain briefly if I am a vector space. (Below you may assume the matrices have real entries.)

(a) $\mathbf{Yes,\ M_3(\mathbb{R})\ is\ a\ real\ vector\ space.}$

(b) $\mathbf{No}$

(c) $\mathbf{Yes}$

(d) $\mathbf{Yes}$

(e) $\mathbf{Yes}$

Answer $\mathbf{(a)\ Yes,\ M_3(\mathbb{R})\ is\ a\ real\ vector\ space.}$ $\mathbf{(b)\ No}$ $\mathbf{(c)\ Yes}$ $\mathbf{(d)\ Yes}$ $\mathbf{(e)\ Yes}$

`mixed` linalg

Problem P4

Am I a (real) vector space (20 pts) ?

(a) $\mathbf{Yes}$

(b) $\mathbf{Yes}, V = \{f \mid f(0) = 0\}$ 是实向量空间

(c) $\mathbf{No}$

(d) $\mathbf{Yes}, V = \{f \mid f(17) = 0\}$ 是实向量空间

(e) $\mathbf{No}$

Answer
$\mathbf{(a)\ Yes}$ $\mathbf{(b)\ Yes}, V = \{f \mid f(0) = 0\}$ 是实向量空间 $\mathbf{(c)\ No}$ $\mathbf{(d)\ Yes}, V = \{f \mid f(17) = 0\}$ 是实向量空间 $\mathbf{(e)\ No}$

`mixed` linalg

Problem P5

Using Julia as a calculator. Include screen shots and written answers. (20 points with -2 for every wrong answer)
(If you get ERROR:UndefVarError: you may have a typo or you may need julia> using LinearAlgebra)(In many cases, copy and paste will just work)

(a) 225

(b) $\|x\|_2 = \sqrt{55}, \quad \|y\|_2 = \sqrt{979}$

(c) $\cos\theta = \frac{225}{\sqrt{55}\sqrt{979}} \approx 0.969, \quad \theta \approx 0.248\ \text{rad} \approx 14.2^\circ$

(d) $x = A^{-1}b$

(e) $b - Ax \approx \mathcal{O}(10^{-16})$

(f) $(b - A * x) \cdot 2^{52} \approx$ 接近整数的数

(g) $A^2 = \begin{pmatrix} 30 & 36 & 42 \\ 66 & 81 & 96 \\ 102 & 126 & 150 \end{pmatrix}, \quad A.^2 = \begin{pmatrix} 1 & 4 & 9 \\ 16 & 25 & 36 \\ 49 & 64 & 81 \end{pmatrix}$

(h) $x = (-0.5\ 0.5\ 0.5)$

Answer

(a) 225 (b) $\|x\|_2 = \sqrt{55}, \quad \|y\|_2 = \sqrt{979}$ (c) $\cos\theta = \frac{225}{\sqrt{55}\sqrt{979}} \approx 0.969, \quad \theta \approx 0.248\ \text{rad} \approx 14.2^\circ$ (d) $x = A^{-1}b$ (

`mixed` unknown